

SEPTEMBER
New Year Edition

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B737 FLEET MAGAZINE



An exclusive fleet magazine for EAL 737 pilots.

Fleet News • Deeper look into B737 incidents and accidents • Breakthroughs in flight • Notices •
System reviews

NOTE

This magazine is for information purposes only. **USE COMPANY AND APPROVED MANUALS AS PRIMARY REFERENCE.**

Thank you for taking time to read The EAL 737 Fleet MAGAZINE. All readers are encouraged to submit their questions and suggestions, regarding this magazine to israelya@ethiopianairlines.com

PREPARED BY

**CAPTAIN HENOK WALE
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September 2025



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MESSAGE FROM CHIEF PILOT



Greetings to you all,

It's hard to believe that three months have already passed since our last newsletter—time truly flies. A lot has happened in our fleet during this period, and I'm pleased to share some exciting updates and developments

1. Fleet Growth & Operational Planning

We've welcomed **three** new B737 MAX aircraft into our fleet, a clear sign of the airline's continued growth. This expansion has required more efficient planning on the scheduling and training fronts to meet increasing demands without compromising safety.

As mentioned in our previous newsletter, senior captains have played a vital role in this effort. At the start of each month, we hold a detailed planning meeting with scheduling and training departments to align crew requirements. This has helped ensure a healthy work-life balance while respecting legal flight hour limitations.

A second meeting mid-month focuses on roster fairness, coordinated by **Capt. Ketema** and **Capt. Tesfatsion**, who will soon hand over responsibilities to **Capt. Hermon** and **Capt. Abraham**.

2. Training & Safety Initiatives

A team led by **Capt. Ayal**, **Capt. Addisu**, and **Capt. Yospph Setarge** is doing excellent work facilitating **route training for new captains**.

Coordination with **maintenance** is being handled by **Capt. Sofonias**, **Capt. Dawit**, and **Capt. Bereket**.

The **safety team**, previously recognized for their outstanding work, is now developing a **tailored chart for ET operations in Mogadishu** and an **APU-on-demand procedure** for ETOPS flights—an initiative that will significantly reduce operational costs.

Continued on next page ...

Efforts to enhance domestic operations are being led by **Capt. Dagmawi A, Capt. Biruk,** and **Capt. Daniel D.**

Preparations for the mock IOSA audit by GHS in August were supported by **Capt. Daniel S, Capt. Tsega K,** and **Capt. Solomon A.**

A pilot satisfaction survey was conducted by **Capt. Daniel S** and **Capt. Samuel K** to identify areas of improvement.

Coordination across these teams was expertly managed by **Capt. Nathan K, Capt. Amanuel,** and **Capt. Dawit FG.**

3.Senior Captain Continuity

We've reached another milestone by ensuring the continuity of the senior captain position. With two senior captains promoted to B777 in August and more expected to transition in September, we successfully recruited 14 new senior captains.

These captains are now paired with experienced mentors and are already contributing fresh ideas to improve operational safety.

Additionally, a new focal point was proposed by **Capt. Luel Hadgu** to liaise with IOCC, and their kickoff meeting was held on September 15th.

4.Upcoming Initiatives

To support the **professional development of our First Officers**, we are launching a **monthly webinar series**. These sessions will feature:

- **Scenario-based discussions**
- **CRM attribute analysis**
- **Experience-sharing segments** led by **senior captains**, including **Capt. Hilina N**, who will guide the legacy-sharing initiative.

Acknowledgments

A special thank you to the team behind this newsletter: **Capt. Israel, Capt. Duresa,** and **Capt. Henok**—your work continues to set a high professional standard.

Closing Notes

We've learned that a **collaborative approach** is key. As the airline grows, coordination becomes more complex, but synergy across departments has proven to be incredibly effective. If we've achieved this much in a short time, imagine what lies ahead if we keep the momentum!

Thank you to all **Captains and First Officers** for your dedication. Appreciation also goes to the **Scheduling, Training,** and **MRO departments** for their continued support.

Congratulations to those who got married, welcomed new babies, or received promotions.

Condolences to those who lost loved ones—our thoughts are with you.

Let us continue to stay safe and support one another.

Warm Regards,
Capt. NATHAN ELIAS
Chief Pilot - B737&B767



FLEET NEWS

Since our last issue, Senior Captains across the 737 fleet have been quietly making a difference in ways that matter most to our daily operations. From solving technical puzzles to easing the workload for fellow pilots, their efforts reflect the teamwork and problem-solving spirit that keeps our fleet moving forward. In this edition, we share a few of the stories and achievements that stood out over the past months as well as a new milestone for our BBJ aircraft.



1.New Approach At MGQ

One of the notable achievements in recent months has been the work of Senior Captains focused on safety, who coordinated closely with the Somali Civil Aviation Authority and Jeppesen to introduce a new approach into Runway 23 at Mogadishu.

Until now, security considerations meant landings were almost always conducted on Runway 05, approaching from the sea. While safe, this procedure often came with the challenge of strong tailwinds. The upcoming release of a tailored approach chart for Runway 23 will provide crews with a safe and secure alternative, improving operational flexibility and safety in Mogadishu.



2.Dire Dawa PAPI Calibration

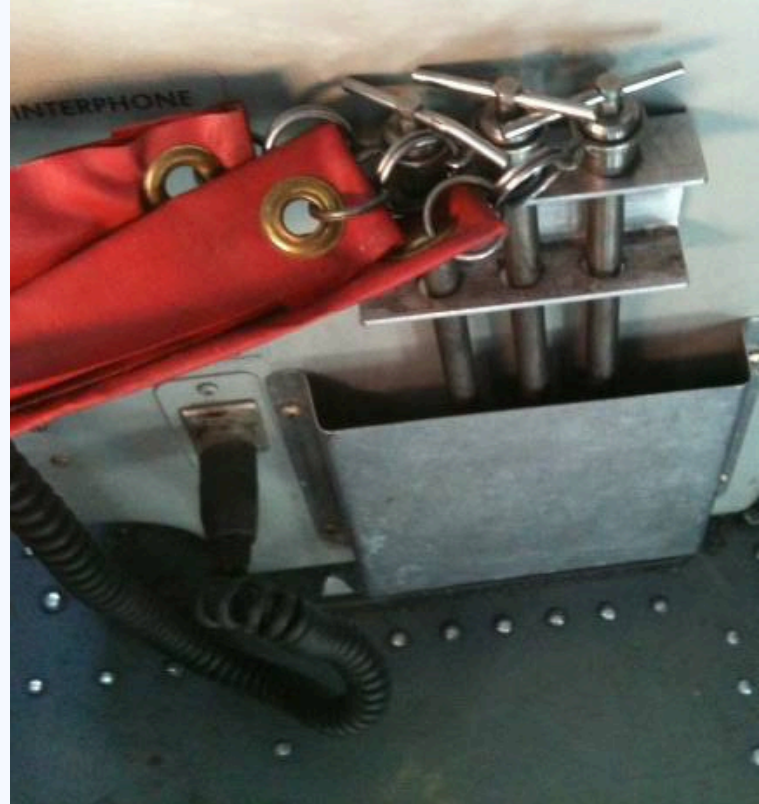
Night operations into Dire Dawa have long tested pilots, with the airport's PAPI system providing unreliable guidance that often made approaches more challenging than they needed to be.

Recognizing the safety implications, Senior Captains worked hand in hand with local air traffic control to address the issue. Their coordination led to the proper calibration of the PAPI, ensuring it now provides accurate information to pilots. The result is a safer, more straightforward approach into Dire Dawa—especially during night flights.

3. Gear Pins Stowage Improvement

In the past, gear pins were stored in the E&E bay, a practice that occasionally led to costly disruptions when pins were overlooked and aircraft had to return after takeoff with the landing gear unable to retract.

To solve this recurring issue, Senior Captains partnered with MRO to rethink the process. The result is a new procedure that keeps gear pins stored in the cockpit, right where they're most visible to the crew. This simple change greatly reduces the chance of pins being forgotten, strengthening both safety and operational reliability.



4. First Business Jet Flight

Alongside the achievements of our Senior Captains, the fleet itself has also welcomed an exciting addition. Our Airline recently acquired a Boeing 737 Business Jet, which completed its first flight not long ago.

The BBJ brings with it a number of unique features that set it apart from our other 737 aircraft. Among these are built-in airstairs, high-altitude landing capability, an icing detection system, three cargo fire extinguisher bottles, a Head-Up Display (HUD) on the captain's side, and an auxiliary fuel system.

These differences enhance both the versatility and safety of the aircraft, while also requiring crews to be familiar with its specific systems. Pilots are encouraged to review the detailed documentation available in Logipad for a full breakdown of the differences and procedures



SAFETY REPORTING

REPORTABLE ITEMS & PROPER E-FORMS USAGE

1.0 INTRODUCTION

The aviation industry is rapidly embracing digitalization, and electronic forms (e-forms) are becoming an essential tool for pilots. They replace traditional paper documentation, streamline reporting, and enhance efficiency and compliance. However, to ensure their effectiveness, pilots must use e-forms properly and consistently.

2.0 Definitions

Occurrences: Accidents, incidents (both serious and minor) and events arising from some element(s) of the aviation system not performing to the expected level.

Accident: is an occurrence associated with the operation of an aircraft which takes place between the time any person boards the aircraft with the intention of flight until such time as all such persons have disembarked.

a) A person is fatally or seriously injured because of:

- being in the aircraft, or
- direct contact with any part of the aircraft, including parts which have become detached from the aircraft, or
- direct exposure to jet blast or when the injuries are to stowaways hiding outside the areas normally available to the passengers and crew; or

b) The aircraft sustains damage or structural failure which:

- Adversely affects the structural strength, performance, or flight characteristics of the aircraft, and
- Require major repair or replacement of the affected component, except for engine failure or damage, when the damage is limited to a single engine

c) the aircraft is missing or is completely inaccessible.

Incident: is any occurrence, other than an accident, that is associated with the operation of an aircraft and affects, or could affect, the safety of operation.

Serious incident: An incident involving circumstances indicating that there was a high probability of an accident and associated with the operation of an aircraft which takes place between the time any person boards the aircraft with the intention of flight until such time as all such persons have disembarked.

Hazard: is defined as a condition or an object with the potential to cause or contribute to an aircraft incident or accident, injuries to personnel, damage to equipment or structures, loss of material, or reduction of ability to perform a prescribed function.

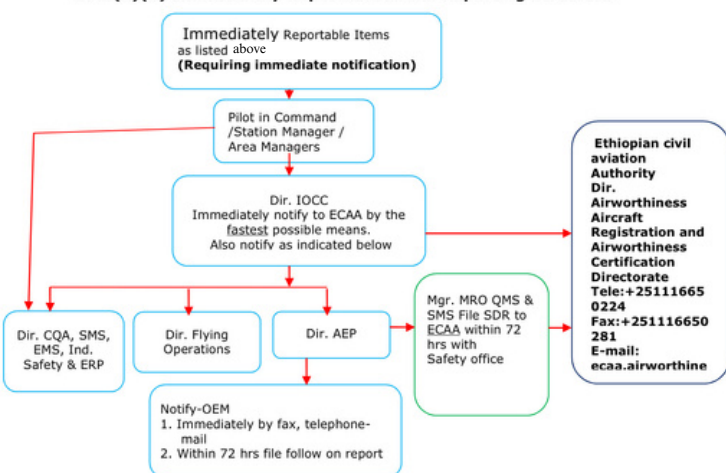
3.0 FOR E.C.A.A

- **Mandatory reportable items-** The failures, malfunctions and defects listed under **EAL SMS Manual section 4.7.3 shall be reported by responsible persons from Flight Operations** to ECAA within 3 days/72 hours after determining that the failure, malfunction or defect required to be reported has occurred.

Immediately Reportable Items

- Primary structural failure
- Control system failure
- Fire in the aircraft
- Engine structure failure or
- Any other condition considered an imminent hazard to safety

4.7.3(ii)(b) Immediately Reportable Items Reporting Structure



Incident Notification

The Pilot in Command (PIC) shall submit report without delay to **ECAA** by filling out appropriate **ECAA forms** of the below incidents.

a)Air Traffic Incident Report- Whenever an aircraft in flight has been endangered by:

1. Near collision with another aircraft or object
2. Faulty air traffic procedures or lack of compliance with applicable procedures by ATC or by the flight crew or
3. A failure of ATC facilities

b)Birds- In the event a bird constitutes an in-flight hazard or an actual bird strike occurs, the PIC shall without delay

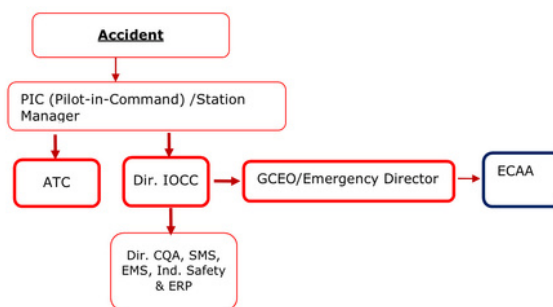
- 1.inform the appropriate ATC and/or ground station whenever a potential bird hazard is observed and
- 2.submit a written bird strike report after landing

c)Dangerous Goods- The PIC shall inform the appropriate ATC facility, if the situation permits, when an in-flight emergency occurs involving dangerous goods on board.

d)Unlawful Interference- The PIC shall submit a report to the local authorities and to the Authority without delay following an act of unlawful interference with the crew members on board an aircraft

Accident Notification

All accidents as defined in the definitions section above shall be reported to ECAA per the following accident reporting structure.



It is advisable to consult chief pilot and/or senior captains for the correct format and type of reporting to ECAA.

4.0 FOR FOPs SAFETY OFFICE

The following report types are expected from pilots to address the reporting system effectiveness.

1. Captains Report

All captain reports are to their respective chief pilot for events related to flight crew, aircraft systems (other than logbook items), ATC, ground services, passengers, weather information and so on as needed.

2. Air Traffic Incident

This form is filed for any occurrences other than an accident intended for the purpose of investigation by the safety office and ECAA.

3. Aircraft Accident/Incident Report

For occurrences that involving serious incidents and accidents in the operation of an aircraft which require full investigation and required by the ECAA as well as the airline corporate safety office. For serious incidents requiring investigation and required by the ECAA as well as the airline corporate safety office.

4. Air Safety Report (ASR)

For events or occurrences during the operation involving aircraft and personnel as AIRPROXY, RA maneuvers, wake turbulence, windshear, technical or non-technical events deemed necessary to be reported intended for investigation and mitigation by the safety office.

5. Bird Strike Report

Required by the safety office and airport Wildlife Management as soon as practicable to devise ways of preventing bird strike incidents/accidents in and around airports for safe operation of aviation activities.

6. FOPs Hazard Report

This is a report available in our system to report any occurrences or events that may result in hazardous situation (health hazard, traffic hazard, etc.) as deemed necessary to be reported to the safety office.

7. Occurrence Report

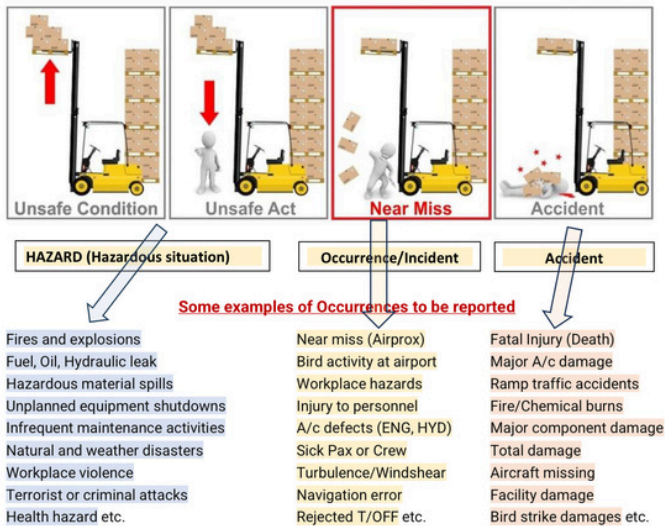
An event or hazardous situation that arise from the activities in the operation area that can be a threat or cause of incident or accident though there is no damage or injury observed.

8. Confidential Hazard Reports

This is an FOP hazard report addressed directly to the safety office designed to encourage reporting of aviation incidents or hazards that have the potential of causing an incident or accident in cases where the person reporting wants to keep his/her details confidential.

9. Voluntary Hazard Reports

This is an FOP hazard report addressed directly to the safety office designed to encourage reporting of aviation incidents or hazards that have the potential of causing an incident or accident in cases where the person reporting is willing to share his/her details



In addition to **captains' report** following reports will suffice the required information.

FOP Hazard Report Confidential Hazard Report Voluntary Hazard Report	Air safety Report (ASR) Air traffic incident report Bird Strike Report Occurrence report	Bird strike report Accident/Incident Report
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5.0 Best Practices

Timely Completion

Fill out e-forms immediately after completing a flight or event to maintain accuracy. Delays in reporting can lead to incomplete or inaccurate records.

Professional Language and Tone

When filling out e-forms, pilots should use clear, concise, and professional language. Avoiding slang, abbreviations (unless standard ICAO/airline codes), and emotional expressions; this ensures the report is easily understood by all recipients.

Compliance with SOPs

Follow company guidance on which forms to complete and when. Ensure forms are routed to the correct department. (FOPs, ECAA etc...).

Use of Remarks Section

Utilize remarks fields to add context when standard fields are insufficient. Keep remarks professional, factual, and concise.

Cross-Checking Before Submission

Before submitting an E-form pilots should:

- Review entries for accuracy.
- Ensure mandatory fields are filled.
- Double-check flight numbers, dates, times (UTC vs local), and airport codes.
- Errors in these fields may cause confusion in flight records or safety databases.

Stay Updated on System Changes

Regularly review company notices or updates regarding e-form usage.



SAFA

Recent Global and Ethiopian Airlines findings.

In the ever-evolving world of aviation, **safety** remains the cornerstone of operational excellence.

Within the European Union and many other global destinations, the **Safety Assessment of Foreign Aircraft (SAFA)** program continues to stand as a rigorous, independent watchdog—conducting thorough inspections to ensure that foreign aircraft meet international safety standards.

From August 30, 2024, to August 29, 2025, SAFA inspectors across various airports logged a range of findings, categorized by severity. These ranged from minor documentation inconsistencies to major technical lapses affecting aircraft systems, cargo handling, and cabin safety.

Most frequent findings were related to defect notification and rectification on Tech log and ADD, flight preparation, Navigation/instrument charts, radio license and preflight inspections.

This article offers a deep dive into key SAFA findings over the last year, with a spotlight on **flight deck**, **technical log issues**, and **specific cases** involving our own airline.

SAFA FINDINGS: GLOBAL OVERVIEW

Category 3 Findings (Major Safety Impact)

1. Refueling with Passengers Onboard

- Catering staff **blocked the main aisle** during refueling, both in the **aft and forward galleys**, restricting emergency egress.
- **No PA announcement** was made to inform passengers during refueling – a mandatory communication requirement.
- Only **one of two exit doors** was open during refueling, despite two passenger stairs being positioned – reducing emergency evacuation options.

2. MEL-Related Noncompliance

- Fuel quantity indicator digits were unserviceable for the left and center tank valves on one aircraft – this was not properly deferred in the MEL.
- Multiple findings revealed **non-compliant rectification methods**. One example includes: **RH Main Landing Gear (MLG) Brake Harness clamp** was broken and replaced with a **safety wire**, with no AMM reference or Boeing authorization for such a fix.

3. Technical Logbook and Maintenance Issues

- Galley waste receptacle access doors were **placarded as inoperative but not taped shut**, posing contamination and fire risks.
- **High-speed tape** was used to secure a belly fairing panel – a method not supported by structural repair manuals.
- Inconsistent documentation of significant damage (e.g., **paint repair on pitot-static port**), with no entries in technical logs.

4. Cargo Handling & Restraint

A cluster of 7 cargo-related findings were noted:

- Cargo not fully secured in all directions – missing nets, straps, and locks.
- Improper restraint materials (e.g., corner ropes not rated for aviation use).
- Pallets exceeding height limitations, risking impact with overhead panels.
- Several locks unsecured, particularly in the aft cargo compartment.

5. Fire Safety

- A hand fire extinguisher near a flight attendant jump seat was not properly secured – one mounting bracket was loose.

Category 2 Findings (Moderate Concern)

1. Documentation and Manuals

- Contradictions in the **OPS Manual** regarding the Fly-Away Kit (FAK) weight: One section states FAK weight must be **added** to DOW/DOI, another says it must be **removed** if not onboard
- The weight of an AKE containing the FAK was listed as **0 kg** – a clear documentation error.
- **Outdated charts** found in the flight deck, including **Jeppesen backups** carried by the Captain on two separate flights.

2. Tech Log Discrepancies

- Use of incorrect MEL interval (**C instead of A**) on ADD.
- Our airline is only approved to extend **C and D** intervals – not A – indicating procedural oversight.

3. Pre-Flight Inspections

- Pre-flight walk-around failed to detect:
- **Flaking tire damage**, with protective cords visible.
- **Missing bolt on engine shroud**.
- **Peeling paint** on critical surfaces initially misclassified under CDL instead of AMM.

Category 1 Findings (Minor but Noteworthy)

1. Flight Deck and Licenses

- Worn **LH pilot seat headrest** reported in poor condition.
- Flight crew failed to **carry valid license or medical certificate** onboard during inspection.
- **Radio licence or certificate of Airworthiness in poor condition**, equipment information unreadable.



MOVING FORWARD: REDUCING FINDINGS, BUILDING TRUST

Consistent SAFA findings – especially those repeating across different locations – underline the need for:

- **Rigorous Flight Deck Checks**
- **Properly Completed Tech Logs**
- **Cabin Readiness Before Boarding/Refueling**
- **Exterior Inspection Thoroughness**
- **Up-to-Date Documentation Onboard**

Every flight is a chance to **demonstrate excellence**, and every inspection is a chance to **learn and improve**.

Let's work together to **minimize findings, preserve our airline's reputation, and ensure passenger safety**.

 **Safety is not just compliance – it's culture.**

2. Unresolved or Incorrectly Logged Defects

- Static discharger noted as missing in logbook (ADD Log) even though **all were installed** at time of inspection.
- **External Aircraft Condition**
- **Loose or missing fasteners** on fairing panels and engine cowlings.
- **Faded or missing placards** on engine and landing gear.
- **Poor paint condition** on fuselage, tail, wings, and engines.

EAL: SPECIFIC FINDINGS

SAFA inspectors recorded **three Category 3 findings in August 2025** related to our operations:

1. Toronto Pearson International (CYYZ) – A350

- Cabin equipment not properly secured.
- Fire extinguisher bracket unsecured

2. WARSZAWA/OKECIE, POLAND (EPWA) – B787-9

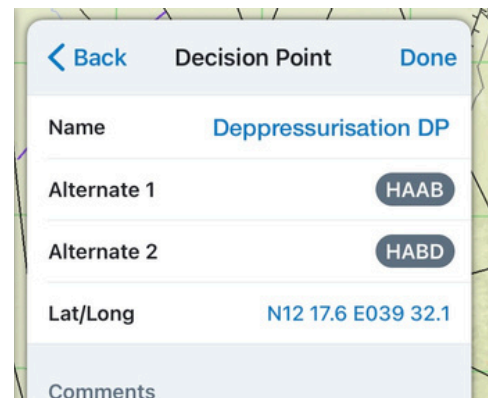
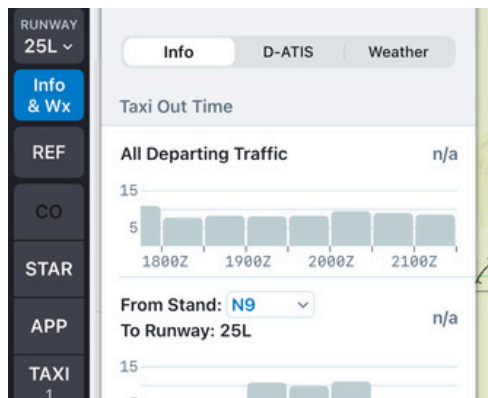
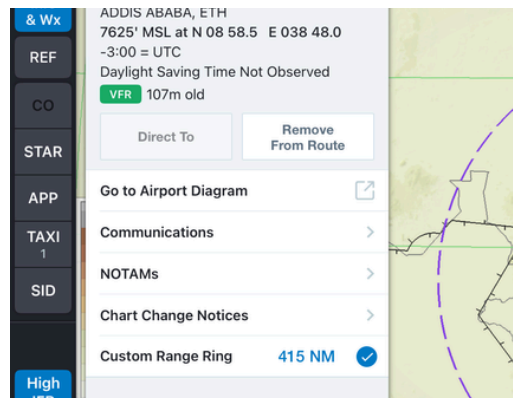
- Seat(s) unserviceable and not identified in cabin log.
- Emergency exit access blocked while passengers were onboard.

3. Rome Fiumicino Airport (LIRF) – B787-9

- Cargo not properly secured and restrained.

FD PRO

Tips and Tricks



1. Custom Range Rings

To increase situational awareness you can use custom range rings. custom range rings can be applied to fixes (including waypoints, navaids, user waypoints, and tailored waypoints), volcanoes, and airports. Can also be used to display EDTO circles.

Creating a Custom Range Ring

To create a custom range ring:

- Tap a fix, volcano, or airport from the enroute map.
- Tap the **Custom Range Ring** option to place a check mark.
- Enter a distance value. This will be the radius of the circle. Ring values are between 5 and 4000 NM.
- Tap **Done**.

2. Taxi Times

To plan for SETO you can use recent taxi time information, along with historical taxi data, that accounts for time of day and year. Results are made more relevant by selecting parking stands or runways.

Displaying Recent Taxi Times

To display recent taxi times for the active flight, first, ensure **Enable Taxi Times** is set to ON in application settings. Then:

1. Tap the **Info and Wx** button.
2. Scroll down in the panel to view Taxi In/Taxi Out Times.
3. Further refine historical time by selecting a runway and parking stand. FliteDeck Pro displays time from the stand to the selected runway.

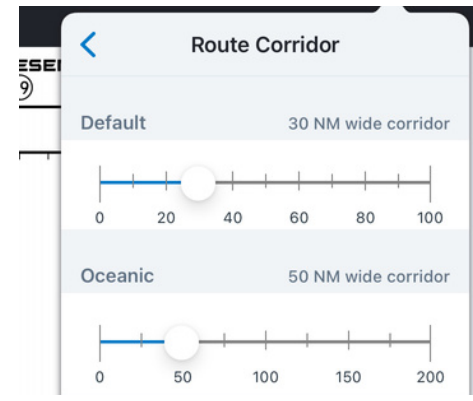
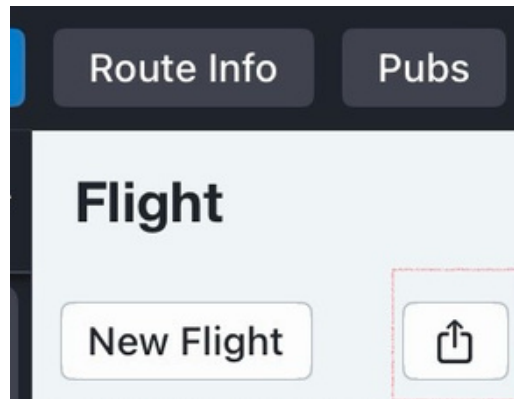
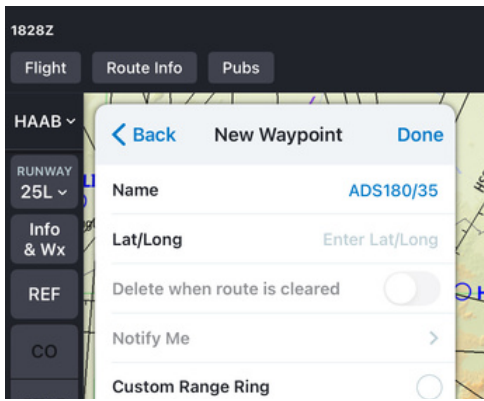
3. Creating Decision Points

They can be used to mark equal time points (ETPs) for ETOPS, or any other critical decision point, such as, diversionary purposes in decompression scenarios.

Creating Decision Points

To create a decision point:

1. Touch and hold the enroute map at the location where you want to insert a decision point. (Alternatively, you can open the Flight Drawer and tap the **Drop a Point** button at the bottom of the drawer).
2. Tap the **Decision Point** flyout and complete the following:
 - To name the marked position, enter a name in the Name field.
 - To associate airport(s), enter airport names in the Alternate fields.
 - To add comments, enter comments in the Comments field.
3. Tap **Done** or away from the popover to dismiss it.



Creating a Radial/DME Waypoint

You can create radial/dme waypoints easily using the simple steps below

To create radial/DME waypoint

1. Long tap the enroute map at the location where you want to insert the waypoint. FliteDeck Pro displays a Location Inspector or Select Point Type popover.
2. Tap Drop a Point at the bottom of the Location Inspector popover or select Waypoint flyout.
3. For the Name field, enter the radial DME in the Name field in this format: navaid, radial/DME (no spaces). For example, for "35 DME on 180 radial off of ADS VOR," you would enter: **ADS180/35** and click Done.

Sharing the Active Flight

To share the active flight:

1. Tap **Flight** to open the Flight Drawer.
2. Tap the **Share** button at the top of the Flight Drawer.
3. From the list of nearby devices, select the device with which you want to share flight info.

NOTE: To share flight information between two devices, the following conditions must be met.

- Both devices are running FliteDeck Pro on the same operating platform.
- Allow Flight Sharing and Local Network settings are set to ON in FliteDeck Pro application settings.
- Bluetooth or Wi-Fi are enabled on both devices and are nearby.

NOTE: Both devices must have the same terminal chart coverage areas to successfully share selected charts.

Adjusting the Route Corridor

The width of the route corridor determines how much information is elevated into the Route List. The wider the corridor, the more information FliteDeck Pro displays in the Route List.

To adjust the width of the route corridor:

1. Tap the **Settings** button.
2. Tap the **Route Corridor** flyout.
3. To increase or decrease the Route Corridor width, drag the Route Corridor slider.



FLAPS

Small surfaces with Big Consequences

1.0 Introduction

Flaps are among the most critical high-lift devices installed on transport category aircraft. Their primary role is to increase lift at lower airspeeds, enabling aircraft to safely take off and land on shorter runways while carrying commercially viable payloads.

The Boeing 737, one of the most widely produced airliners in history, employs a sophisticated combination of leading edge Krueger flaps and slats together with double-slotted trailing edge flaps.

This configuration enables safe operations across diverse conditions – from short runways to heavy-weight takeoffs in hot-and-high airports. However, because of their mechanical and aerodynamic complexity, flaps are also a source of potential abnormal and emergency scenarios.

2.0 Aerodynamic Principles of Flaps

At their core, flaps work by increasing the camber and effective wing area, which changes the lift coefficient of the wing. By extending flaps, the maximum lift coefficient (CL_{max}) rises significantly, reducing stall speed. This is why approach speeds with full flaps are often 20–40 knots slower than with a clean wing configuration.

However, flap extension also introduces side effects:

- **Drag increases**
- **Nose-Down Pitching moment**
- **Longitudinal Stability Issues**

Aerodynamically, flaps shift the lift-drag polar curve, making high lift possible at low speeds but at the expense of efficiency.

For landing, this trade-off is desirable, as drag helps steepen the approach path and reduce rollout distance.

For takeoff, moderate flap settings provide additional lift while limiting drag to preserve climb gradient.

The 737's multi-slotted Fowler flaps are particularly effective. They not only rotate downward but also slide rearward, significantly increasing wing area and maintaining attached airflow through slot effects.

Leading edge devices such as Krueger flaps deploy forward from the wing undersurface, smoothing airflow and delaying stall at high angles of attack.

Types of Flaps

Flap design varies widely depending on aircraft type, but four main trailing edge types exist:

1. Plain Flap
2. Split Flap
3. Slotted Flap
4. Fowler Flap

In addition, leading edge devices provide crucial stall margin protection:

Krueger Flaps: Panels that rotate from the wing's undersurface. Common on Boeing aircraft, including the B737.

Slats: Move forward to create a slot that re-energizes airflow over the wing's upper surface.

The Boeing 737 incorporates a hybrid design:

Trailing Edge: Double-slotted Fowler flaps.

Leading Edge: Krueger flaps inboard, slats outboard.

This combination maximizes low-speed performance without excessive drag, making the 737 versatile for short and medium-length runways.

Performance Effects on the 737

The choice of flap setting directly affects performance in both takeoff and landing.

Takeoff: Flaps 1 or 5 are common for normal operations, balancing climb gradient and noise requirements. Flaps 10 or 15 may be used for short runways or high-weight operations where extra lift is required.

Trade-off: higher flap settings reduce takeoff speed but penalize climb performance.

Landing: Flaps 30 is the most frequently used setting, providing a balance between drag, go-around thrust margin, and landing distance. Flaps 40 may be selected for short or contaminated runways, but requires higher thrust in go-around.

Performance tables account for flap position in calculating V-speeds and landing distance required (LDR). Pilots must strictly adhere to these tables, as incorrect flap selection can compromise safety margins.

Go-around performance: lower flap settings improve acceleration and climb capability, but increase approach speed and landing distance. Thus, operational judgment is required.

Operational Considerations

Flap usage requires several operational strategies:

Noise Abatement: Lower flap settings reduce drag, allowing quieter climbs, especially near noise-sensitive airports.

Fuel Burn: Less drag during approach translates into lower fuel consumption if flaps are extended later in the descent.

Adverse Weather: In gusty or contaminated conditions, higher flap settings may be avoided to reduce the risk of a tail strike or loss of control.

Crosswinds: Large flap deflections can increase lateral instability. Crews may use Flaps 30 instead of 40 to maintain better handling in strong crosswinds.

Windshear: Takeoff flap settings must balance low-speed performance with maximum energy reserve. During approach, too much drag could worsen effects of windshear.

Non-Normals

Flap malfunctions are among the most significant non normal conditions a B737 pilot may encounter.

Causes that might lead to flap malfunctions:

- Hydraulic system issues
- Mechanical failures
- Asymmetry detection and protection activation
- Position sensing faults
- Electrical failures
- Structural or foreign object interference
- Crew induced or operational factors (exceeding placard speeds)

Common flap non normals on the B737

- Leading edge flaps transit
- Trailing edge flap asymmetry
- Trailing edge flap disagree

Crew should consider the following:

- Check and double check your landing performance
- Select the most suitable airport and runway for landing
- Reduce airplane gross weight (burn off fuel)
- Fly a wide pattern
- Plan for a slower flap extension if using alternate flap extension
- Establish on long final (approximately 10 NM) to reduce any excess speed early
- Use ILS approach if available
- Touch down on the normal touchdown zone
- Use reverse thrust immediately to the fullest
- Use of autobrake is recommended

Reporting: any flap abnormality must be properly documented in the maintenance logbook. Flight crews are also encouraged to file voluntary safety reports if abnormal flap handling or errors occur, supporting a strong safety culture.

Crew Resource Management (CRM): Flap non-normals highlight the importance of clear communication, precise checklist discipline, and workload management. Approaches with abnormal flap settings demand heightened situational awareness, especially regarding approach speed, runway length, and go-around viability



WINDSHEAR

THE HIDDEN DANGER

1. Introduction

Windshear is an invisible but potentially catastrophic hazard to aviation safety. It poses the greatest threat during takeoff and landing when aircraft operates close to the ground with limited performance margins.

Over the years, several major accidents have underscored the lethal potential of windshear. Despite advances in detection systems, meteorological tools, and simulator training, the phenomenon continues to challenge pilots worldwide. windshear is “an invisible threat to pilots” that requires constant vigilance.

2. Definition

Windshear refers to a rapid change in wind velocity (speed or direction) over a relatively short distance in the atmosphere. It can occur in both horizontal and vertical planes:

Horizontal windshear: sudden shifts in wind direction or speed along the aircraft’s flight path.

Vertical windshear: abrupt changes in wind strength with altitude.

Although windshear can occur at any altitude, its most critical impact is at low levels, particularly near airports.

Meteorological conditions that commonly generate windshear:

- Thunderstorms & associated downdrafts.
- Frontal zones, especially cold fronts.
- Temperature inversions with strong upper-level winds.
- Microbursts, the most hazardous form of localized windshear.

3. Effects on Aircraft Performance

The primary danger of windshear lies in its impact on aircraft energy management during low-level operations.

On approach: A sudden loss of headwind reduces indicated airspeed and lift, causing an uncommanded descent or undershoot of the runway.

On takeoff: A sudden increase in tailwind may reduce acceleration, increase required runway length, or degrade climb performance.

Vertical shear: Downdrafts can overwhelm climb capability, forcing the aircraft toward terrain.

Because takeoff and landing involve aircraft operating close to stall speed and in high-drag configurations, there is little margin for error. Small wind changes can have disproportionately large effects on performance.

4. Recognition of Windshear

Timely recognition is critical. Windshear can be detected through:

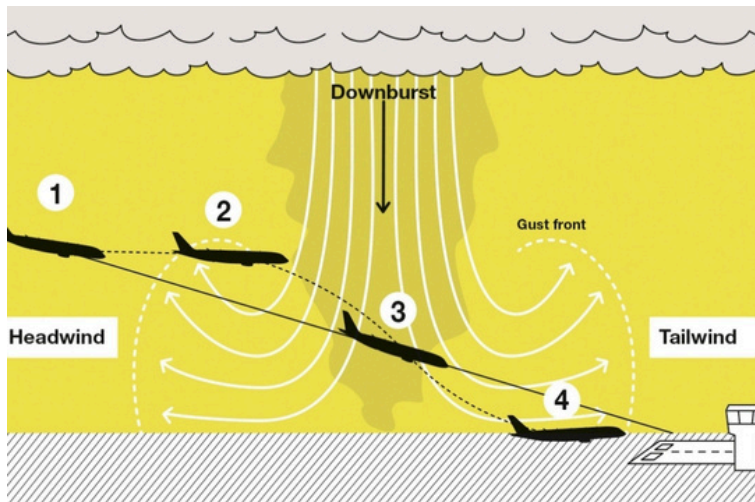
- **Visual-cues:** Rain shafts, virga (rain that evaporates before reaching the ground), or localized dust, diverging surface winds near thunderstorms.
- **Cockpit-cues:** Rapid fluctuations in airspeed, groundspeed, or vertical speed, Unusual pitch or thrust requirements, Stall warnings or stick shaker activations despite proper configurations.
- **Technological cues:**

1. **Predictive windshear systems (PWS):** weather radar-based tools that scan ahead for hazardous conditions.

2. **Reactive windshear systems (RWS):** sensors that monitor aircraft performance and issue alerts when the aircraft is actually in windshear.

5. Microbursts

Microbursts represent the most dangerous form of windshear. They are intense, localized downdrafts, often less than 4 km in diameter that strike the ground and spread outward in all directions.



A typical microburst encounter sequence:

1. Aircraft initially experiences an increasing headwind, giving a false sense of excess lift.
2. As the downdraft is entered, rapid descent rates overwhelm climb performance.
3. Upon exiting, a strong tailwind causes further airspeed loss, leaving the aircraft with minimal energy near the ground.

This deadly sequence has been implicated in several accidents, most notably Delta Flight 191 at DFW in 1985. Since then, microbursts have been the focus of improved detection and training initiatives.

6. Avoidance Strategies

The most effective windshear defense is AVOIDANCE

Preflight preparation:

- Review forecasts for thunderstorms, gust fronts, and unstable weather.
- Note airport-specific LLWAS (Low Level Windshear Alert System) availability.

In-flight strategies:

- Use on-board weather radar to identify convective cells.
- Heed ATC warnings and PIREPs of windshear activity.
- Stay clear of thunderstorm cells and heavy precipitation. If the presence of windshear is confirmed, delay takeoff or do not continue an approach.

7. Precautionary Measures

If windshear is suspected, be alert to the danger signals and be prepared for the possibility of an encounter.

During takeoff:

- Plan to use the longest suitable runway provided it is clear of areas of known windshear.
- Full takeoff thrust is recommended Unless a dispatch performance requirement restricts it.
- Use an appropriate takeoff flap setting. Flaps 5,10 or 15 recommended for optimum performance.
- Consider increasing rotation speed in accordance with approved performance manual

Crew coordination and awareness is very important!

During approach:

- Use the most suitable runway considering factors like tailwind and crosswind limitations and of course one that is clear of suspected windshear
- Use an appropriate landing flap setting. Flaps 30 for B737.
- Be stabilized on approach no lower than 1000 ft to improve windshear recognition
- If the autothrottle is planned to be disengaged for landing add an appropriate airspeed correction
- Avoid large thrust changes in response to sudden airspeed change

Again crew coordination and awareness is very important!

8. Recovery Techniques

When avoidance fails, disciplined recovery is the last line of defense.

Decisive, trained responses are critical.

Hesitation or deviation from procedure often worsens the outcome.

Accomplish the windshear escape maneuver found in the 737 QRH.

9. Operational Considerations

Operators incorporate windshear procedures into training, SOPs, and CRM principles. Key practices include:

- Simulator training with realistic windshear scenarios.
- Standardized callouts and division of duties during recovery.
- Operational discipline, including go-around or diversion decisions when conditions warrant.

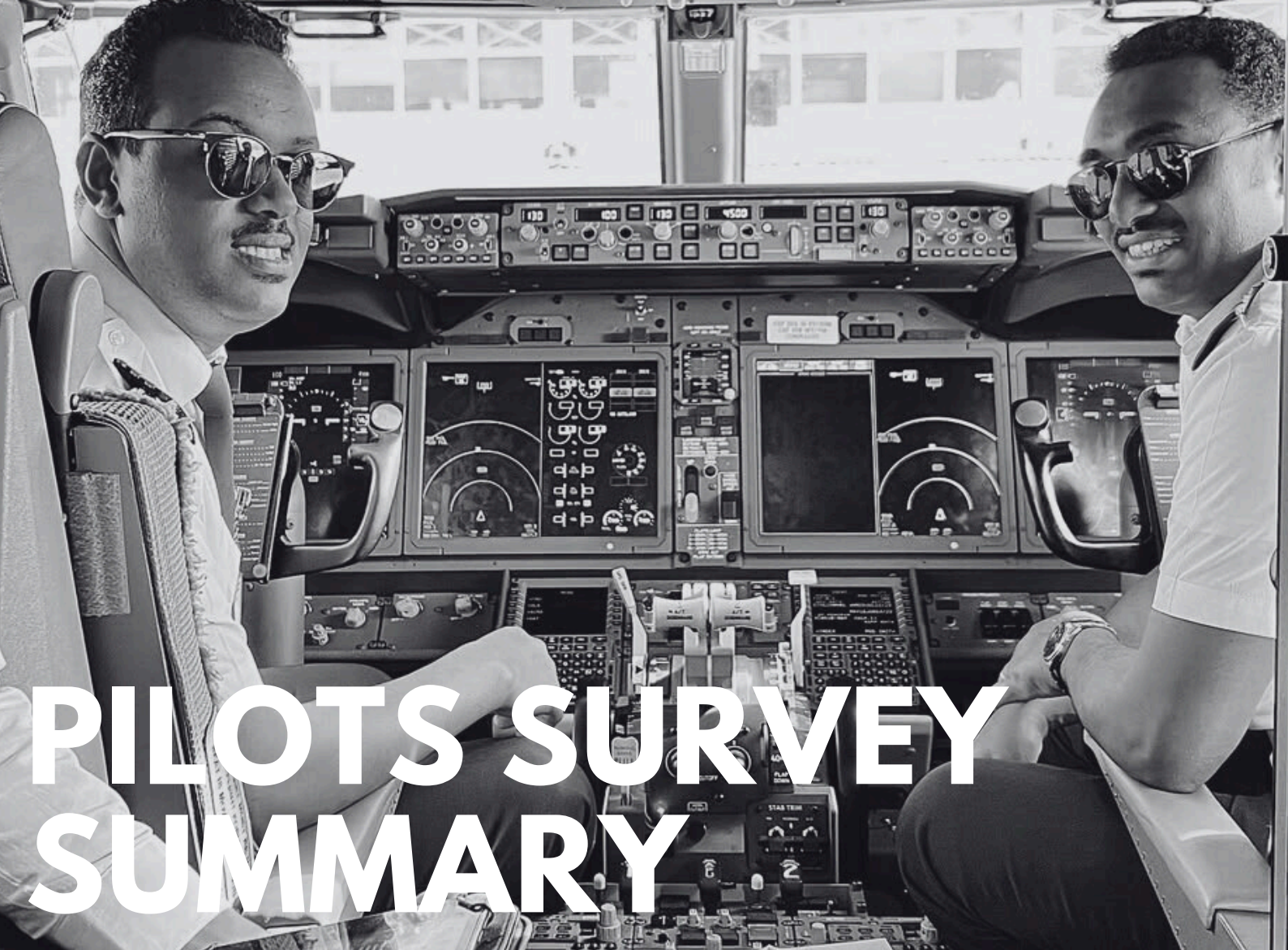
survival often depends not just on technical skill but on the crew's willingness to reject unsafe approaches.

10. Conclusion

Windshear is a unique threat: invisible, unpredictable, and potentially overwhelming. Yet with the right combination of awareness, avoidance, recognition, recovery discipline, and reporting, pilots can effectively manage the risk.

Technological advances, such as the predictive and reactive systems on modern aircraft have greatly enhanced safety.

However, as all references emphasize, the ultimate safeguard remains the **alert, trained, and disciplined flight crew.**



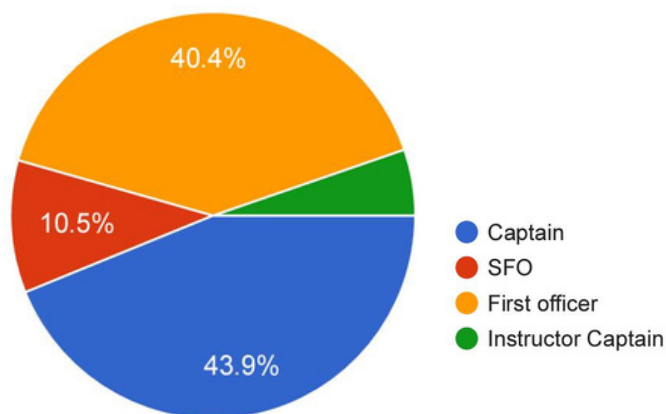
PILOTS SURVEY SUMMARY

In our ongoing effort to ensure that every pilot's voice is heard, a new Boeing 737 Pilot Satisfaction Survey was conducted, gathering input from more than one hundred pilots across our fleet.

The results provide a valuable snapshot of how crews perceive recent operational changes, the role of Senior Captains, and areas where we can continue to improve.

Who Responded

Among **114 respondents**, the group was almost evenly split between **Captains (40.4%)** and **First Officers (43.9%)**, with **Senior First Officers** making up **10.5%** and **Instructor Captains 5.2%**. Half of the respondents have been with Ethiopian Airlines between five and ten years, while the rest(46.5%) have been with the airline for less than five years. This shows a strong representation of both experienced and relatively new pilots, giving the survey a balanced perspective.



Engagement with Senior Captains

Just over half of the pilots (54.8%) reported not working directly with Senior Captains yet, which reflects the relatively recent introduction of the role. However, among those who had, most gave positive feedback. Senior Captains were rated 3.79 out of 5 on improving scheduling fairness, with nearly 43% rating them "4" and a quarter giving the highest possible score.

On fatigue management, two-thirds of pilots said they had not yet observed improvements, but about a quarter (26.3%) noted positive changes. A similar pattern appeared in communication with maintenance, where 30.4% felt improvements while 64.3% had not seen a difference yet. Still, one-third of pilots felt that Senior Captains' involvement had led to quicker resolution of technical issues.

Safety and Culture

One of the most encouraging results came in the area of safety culture. A strong 75.7% of pilots reported clear improvements in how questions and issues were addressed, and Senior Captains received an average rating of 4.21 out of 5 for promoting a positive safety culture. Nearly 70% observed improvements in how safety concerns are escalated, and 79.6% described Senior Captains as approachable and receptive to feedback.

When asked about adherence to SOPs, most pilots (52.2%) agreed that Senior Captains help ensure compliance, and nearly a third (31.3%) said they actively encourage crews to contribute input for SOP updates based on practical experience.

Areas for Improvement

The open comments highlighted constructive suggestions. Several pilots mentioned crew scheduling fairness as an area where further work is needed, noting that some still face more minimum-rest sequences than others.

Others pointed to the need for better coordination with maintenance, stronger rostering practices, and even attention to ground-side issues such as crew transport and hotel facilities.

A few also raised the importance of supporting mental health and wellness, reflecting growing awareness of workload impacts on well-being.

Overall Impact

Despite some areas of growth, the overall impact of Senior Captains was rated highly: 58.3% of respondents called their role "effective," and another 8.7% said "very effective." Importantly, more than 92% of pilots recommended continuing Senior Captain involvement in operational improvements, showing broad support for the program.

As for the future, two-thirds of respondents said they might consider applying for the Senior Captain role themselves someday, while 24.1% expressed strong interest. This indicates that the position is seen not only as useful but also as aspirational within the pilot community.

Conclusion

The survey paints a picture of a program still in its early stages, but already making visible progress— especially in strengthening safety culture and providing new channels of communication.

While there is more to do in the areas of scheduling, fatigue management, and operational coordination, the overwhelming support from pilots suggests that the Senior Captain initiative is a step in the right direction.

With continued feedback and collaboration, it has the potential to reshape our fleet culture in ways that benefit every pilot.

Thank You